And indeed, this value is crucial to the `split` method, because each of the two partitions created will need to have at least m points: therefore, since we are using a single index split[12], the possible values for this split index goes from `m` to `M-m`. In our example, as shown in the middle section of figure 10.18, this means:

- If `m==2`, then we can choose any index between 2 and 6 (5 choices);
- If `m==3`, then the alternatives are between 3 and 5 (3 choices).

Now suppose we had chosen `m==3`: the bottom section of figure 10.18 shows the resulting split for each of the three alternative choices we have for the split index; we will have to choose the one that minimizes variance for both nodes (usually, we minimize the sum of the variances), but don't be mistaken: we only minimize variance along the direction we perform the split, so in the example we will only compute the variance of the y coordinates of the two sets of points. Differently from R-trees we won't try to minimize bounding envelopes overlap at this stage, although it turns out that reducing variance along the direction that had the highest variance brings us, as an indirect consequence and to some extent, a reduction of the average overlap of the new nodes.

But also, as we will see, with Ss$^+$-trees we will tackle the issue of overlapping bounding envelopes separately.

For now, to finish with the insert method, please take a look at listing 10.14 for an implementation of the `minVarianceSplit` method: as mentioned, it's just a linear